

CHEM 101 Topic 10: Chemical Bonds — Answer Key

Section 1: Ionic and Covalent Bonds

Problem 1: Ionic bond. Reasoning: K is a metal and O is a nonmetal $\rightarrow$ electron transfer.

Problem 2: NaF has a much higher melting point and is more soluble in water than NH_3 . Reasoning: Ionic bonds are stronger than the intermolecular forces between covalent molecules.

Problem 3: A molecule is a distinct group of atoms; a crystal lattice is a repeating 3D grid. Ionic bonds form the lattice.

Section 2: Electron Dot Structures

Problem 4: 26 valence electrons. ($\text{P}:5 + 3 \times \text{Cl}:7$).

Problem 5: S in center, two single bonds to H, and two lone pairs on S.

Problem 6: $\text{H}-\text{C}=\text{N}$: (with triple bond $\text{C}=\text{N}$ and one lone pair on N).

Section 3: Molecular Geometry (VSEPR)

Problem 7: Linear.

Problem 8: Trigonal Pyramidal.

Problem 9: Tetrahedral.

Section 4: Coordinate Bonds & Polarity

Problem 10: Coordinate covalent: one atom donates both electrons. Standard: each atom contributes one.

Problem 11: Polar covalent. N is more electronegative than H.

Problem 12: Polar. The Bent geometry prevents the polar bonds from canceling out.